Perspective Homography

A Robot's Journey from Map to Reality

Team 12: Kaiwen Xu, Yudian Ke, He huang, Levis Li



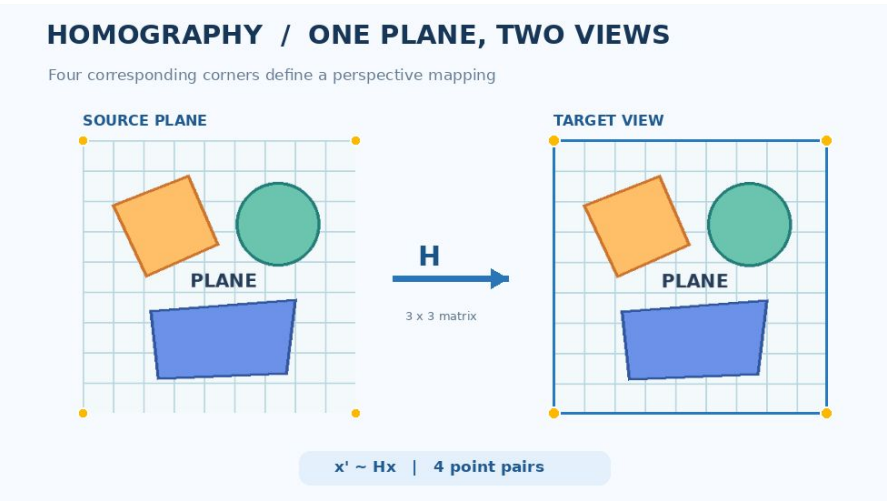
Table of Contents

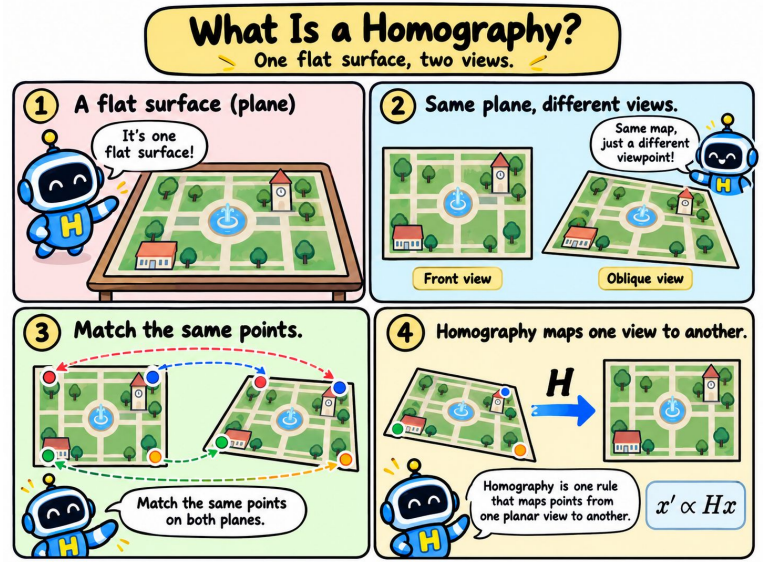


1. Core Concepts
2. Overview
3. Why the Map Looks Skewed
4. How Perspective Forms
5. One Plane, Two Views
6. What Changes, What Stays
7. Finding the Mapping
8. Inside the Matrix
9. Rectification: Bring the Map Back
10. Powerful , with Limits
11. When Matches Get Messy
12. Beyond the Flat Map

Core Concepts

- **Definition:** A homography is a **projective transformation** between two planes.
- **Equation:** $x' \propto Hx$
  H: 3x3 homography matrix
- **Applied on: planar scenes/objects** or two views of the same plane.
- **Minimum requirement:** At least **4 point correspondences** are needed to estimate **H**.
- **Key Property:** Homography preserves **straight lines and collinearity**, but generally does **not preserve lengths, angles, or parallelism**.
- **Common Applications: Perspective rectification, panorama stitching, planar tracking, and pose estimation.**





OVERVIEW

How Homography Works

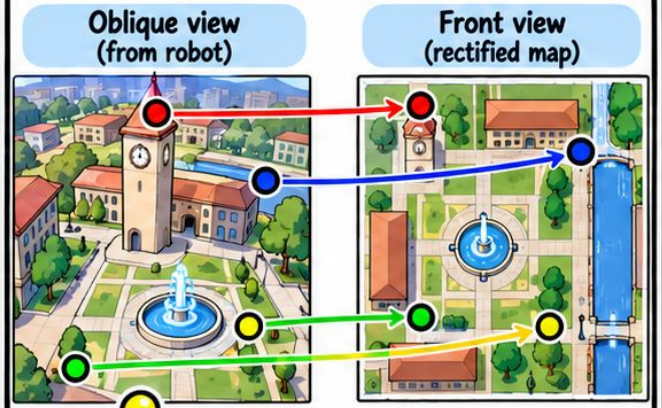
1. Understand the Perspective



A flat plane looks distorted from a different viewpoint.

VIEWPOINT

2. Match Corresponding Points



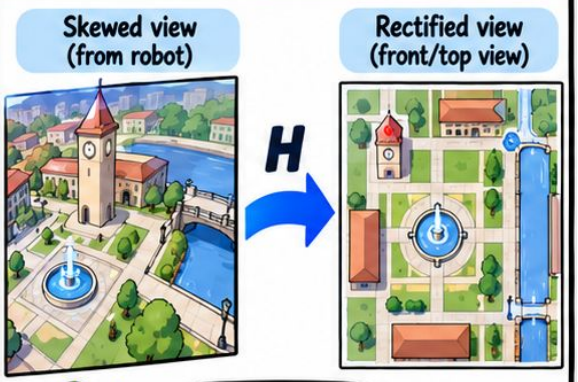


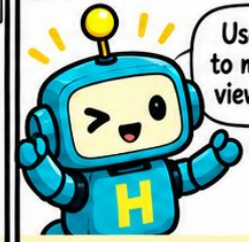
Match the same physical points in both views.

$x' \sim Hx$

MATCHING

3. Rectify the View





Use homography to map one planar view into another.

Works best for one plane.



RECTIFICATION

Observe → **Match** → **Transform**

Why Maps Look Skewed

1 / A Map, Two Views

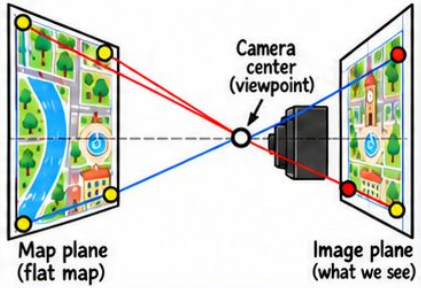
1 Planning a trip to the town!



2 The same flat map looks tilted



3 Follow the rays



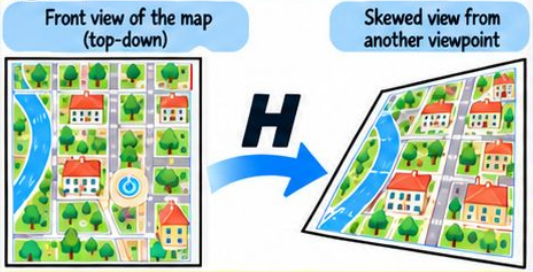
Light rays from the flat map form a perspective view, turning the 3D scene into a 2D image.

4 Rotating or resizing is not enough



Rotation and uniform scaling cannot remove the perspective. The edges still converge!

5 Meet the homography

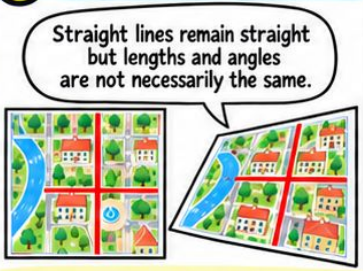


A homography maps one view of a plane to another.
It is a single rule (a 3×3 matrix H) that links the points on this plane.

6 Same map, different views in the real world

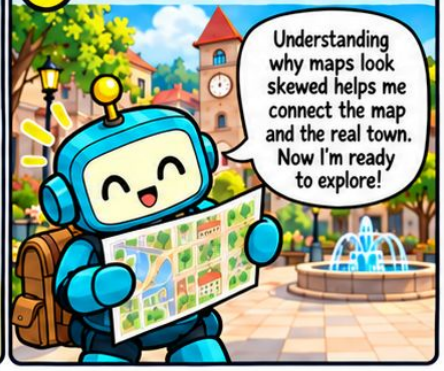


7 Key idea: what is preserved?



On a plane, a homography preserves straight lines, but it does not preserve distances or angles.

8 Now I can explore with confidence!



PERSPECTIVE HOMOGRAPHY

FOLLOW THE LIGHT

2 / How perspective forms

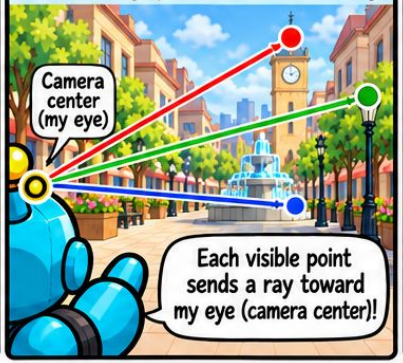
1. Arrive in town!



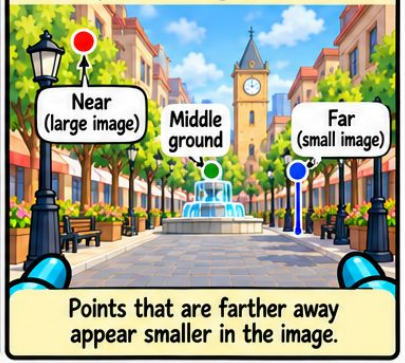
2. Looking down the street



3. Every point sends a ray



4. Depth changes the size



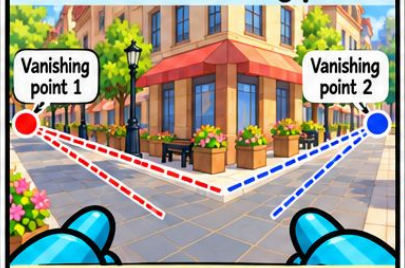
Points that are farther away appear smaller in the image.

5. Parallel lines meet



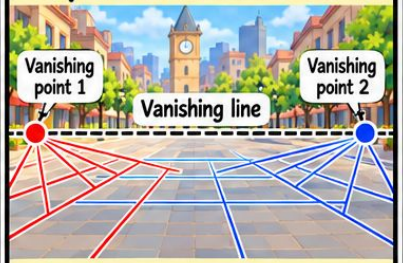
The parallel edges of the road look like they meet at a vanishing point in the distance!

6. A different direction, another vanishing point



Looking in a different direction, another set of parallel lines meet at a different vanishing point.

7. For one plane, vanishing points lie on a line



For a single flat plane (like the ground), the vanishing points for different directions lie on the same vanishing line.

8. Following the light reveals perspective



Perspective is the result of projecting the 3D world onto an image plane along straight rays of light!

PERSPECTIVE HOMOGRAPHY

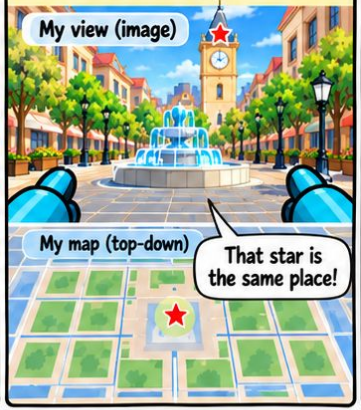
ONE PLANE, TWO VIEWS

3 / What does a homography connect?

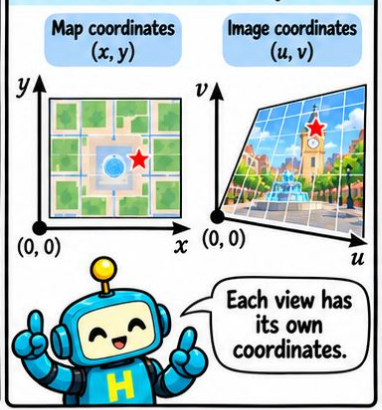
1. Walking through town



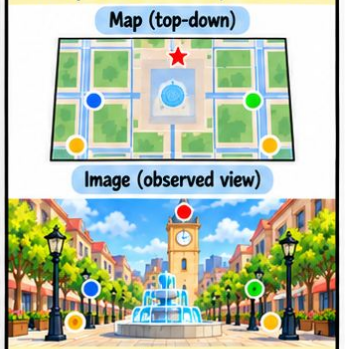
2. The same landmark



3. Two coordinate systems

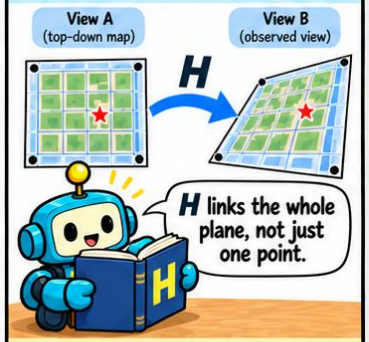


4. A point correspondence



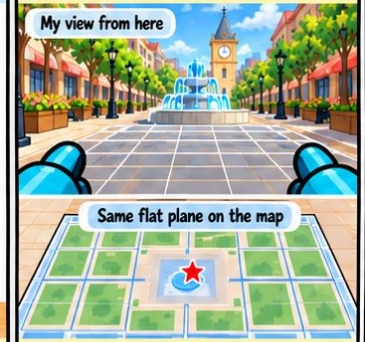
Corresponding points represent the same physical locations on the plane.

5. One rule for the plane



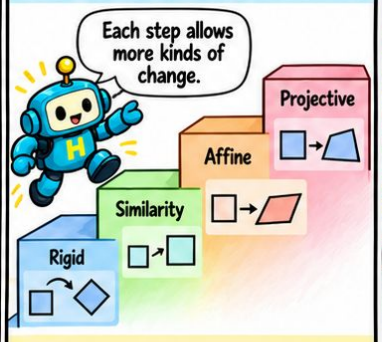
A homography is an invertible projective mapping of a plane.

6. Still the same plane



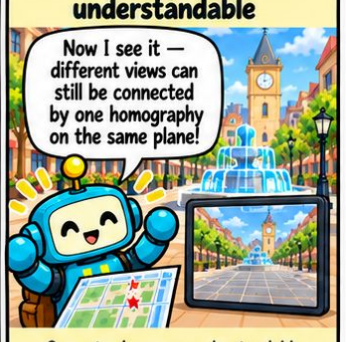
Different viewpoint, but it is still the same flat plane in the real world.

7. A ladder of transformations



Rigid: move and rotate. Similarity: scale too.
Affine: keep parallels. Projective: allow convergence.

8. Geometry becomes understandable



Geometry becomes understandable because perspective connects two views of the same plane!

PERSPECTIVE HOMOGRAPHY

WHAT CHANGES? WHAT STAYS?

4 / The rules of projective geometry

1. Exploring a new street in town!

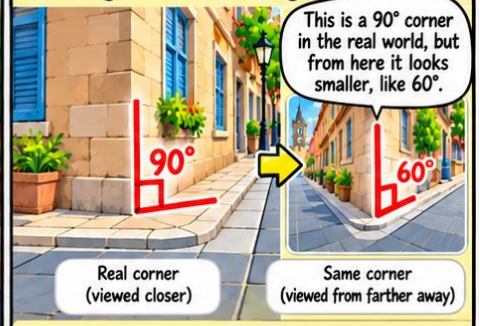


2. Straight stays straight



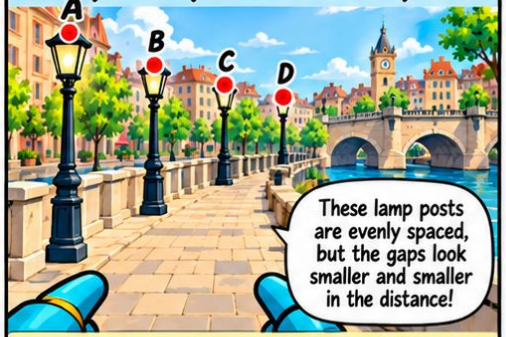
Straight lines, collinearity, and incidence are preserved by a homography.

3. Angles can change



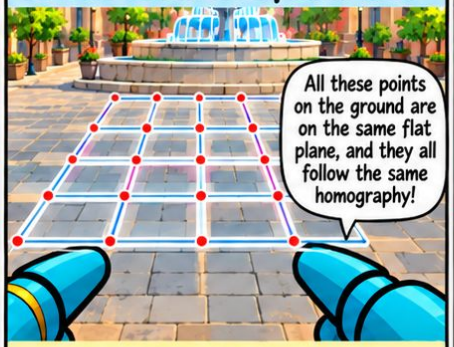
Angles, lengths, areas, and parallelism are not generally preserved.

4. Equal steps can look unequal



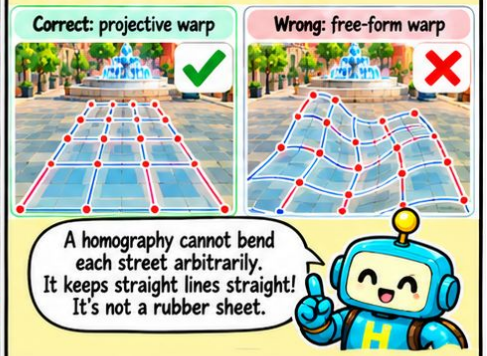
Equal steps on the same line (or plane) can look unequal in perspective.
Midpoints and ordinary distance ratios need not stay the same.

5. One consistent planar rule



A single homography links the whole plane — the same rule for every point on the plane.

6. Not a rubber sheet



A homography is a rigid projective mapping of the plane with one consistent rule. It is not a free-form warp.

PERSPECTIVE HOMOGRAPHY

FINDING THE MAPPING

5 / Four Pairs, One Rule

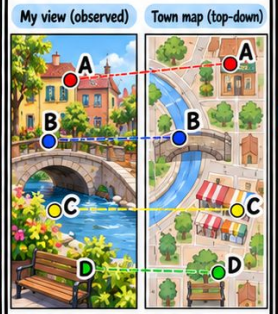
1 A wrong turn in town



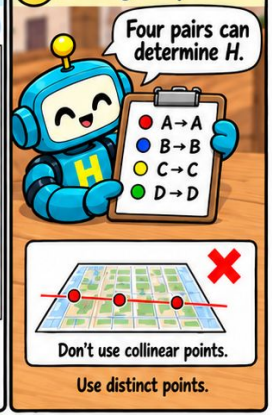
2 Take out the map



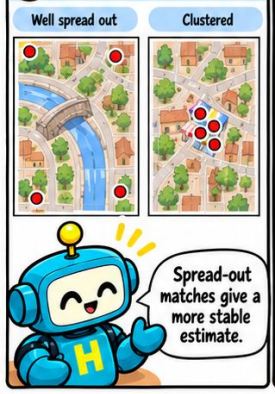
3 Match the same places



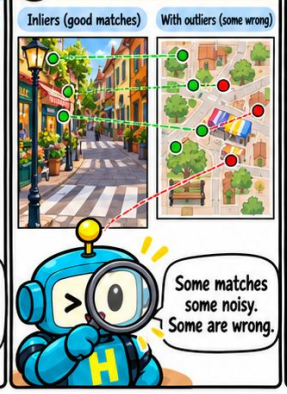
4 Four good pairs



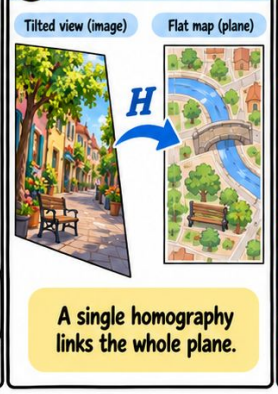
5 Spread out the evidence



6 Real matches are imperfect



7 One rule for one plane



8 Ready for the next step

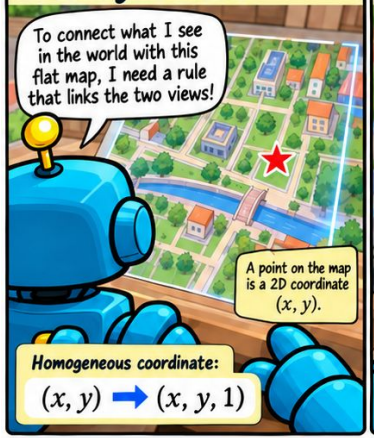


PERSPECTIVE HOMOGRAPHY

INSIDE THE MATRIX

6 / Coordinates, H, and eight degrees of freedom

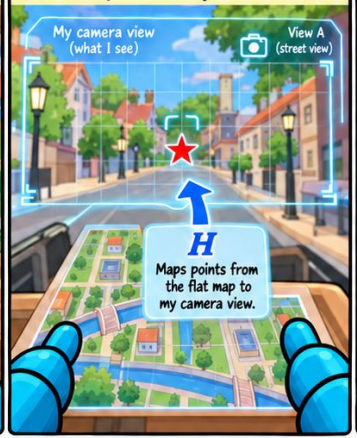
1. Analysis mode



Homogeneous coordinate:

$(x, y) \rightarrow (x, y, 1)$

2. H maps the map to what I see



3. Apply H, then divide by scale

1 Pick a point on the map. In homogeneous form:

$(x, y, 1)$



2 Apply the 3 × 3 matrix H.

$(x', y', w') = H(x, y, 1)$

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

3 Divide by w' to get the visible 2D point.

$$u = \frac{x'}{w'} \qquad v = \frac{y'}{w'}$$



The extra 1 lets H include perspective effects, and dividing by w' gives the final 2D location!

4. One matrix, one rule



$$x' \sim Hx$$

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

$\sim$ means equal up to scale.

H has 8 degrees of freedom because a 3 × 3 matrix (9 numbers) is unchanged if we multiply all entries by a nonzero scale.

5. Multiply, then divide



I multiply by H to get (a, b, w'), then divide by w' to find where the point appears in my view!

6. Reverse or combine

I can undo a mapping, or chain two mappings together to plan my route!

Reverse the mapping
(go from my view back to the map)



Combine two mappings
(go from view A to view C through view B)



Use H^{-1} to go back, or combine two mappings: the result is H_2H_1.

PERSPECTIVE HOMOGRAPHY

BRING THE MAP BACK

7 / Rectification and image resampling

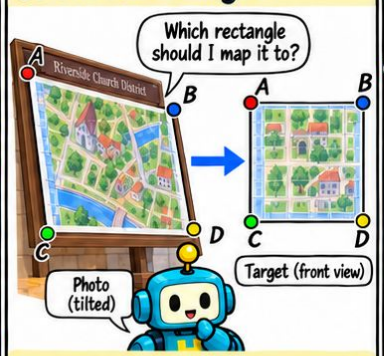
1. A new map in town



2. Fix the perspective

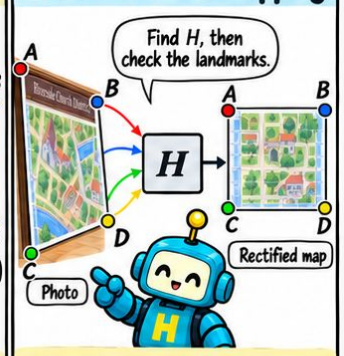


3. Choose a target



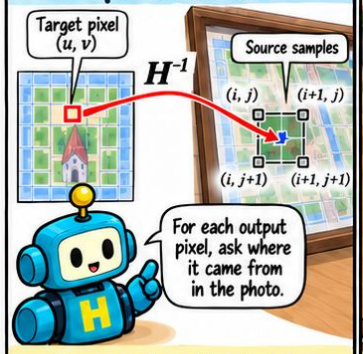
Use a rectangle with a known size or aspect ratio to recover meaningful proportions.

4. Estimate the mapping



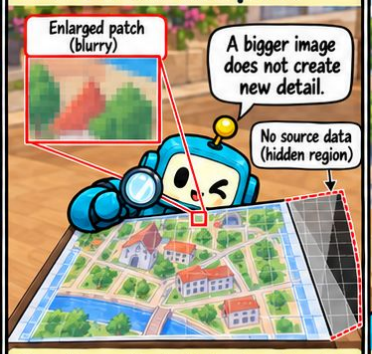
Turning a quadrilateral into a rectangle alone does not reveal its true aspect ratio.

5. Sample backward



Inverse mapping avoids scatter holes; interpolation estimates colors between samples.

6. Read the fine print



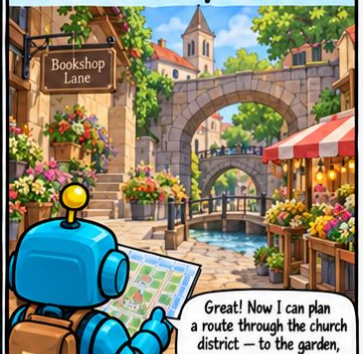
Resampling can soften details. Cropping and boundaries matter; hidden content stays missing.

7. Rectified map result



The rectified map aligns with my map and reveals new streets, landmarks, and connections!

8. Time to explore!



PERSPECTIVE HOMOGRAPHY

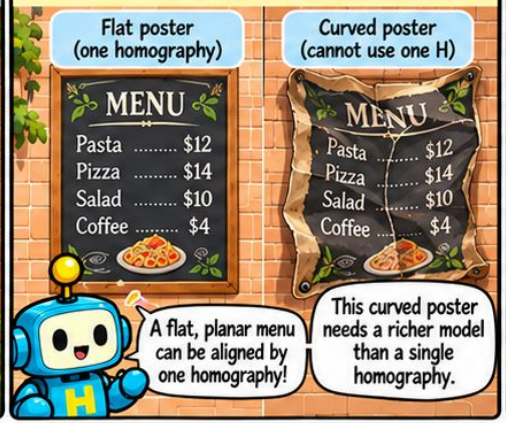
POWERFUL, WITH LIMITS

8 / Assumptions, failures, and applications

1. A hungry stop



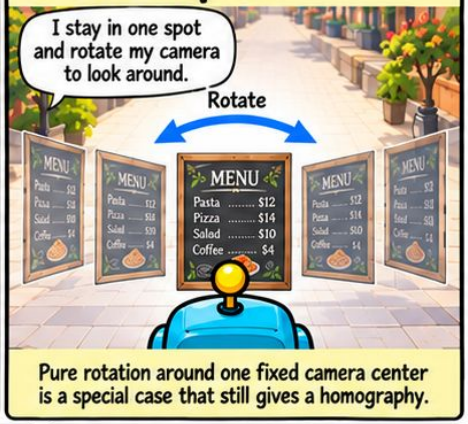
2. Flat works; curved does not



3. Step sideways

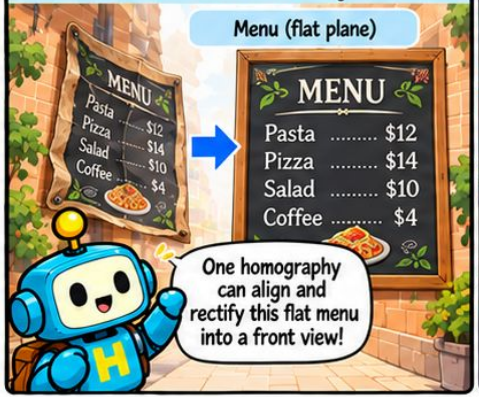


4. Turn in place



Pure rotation around one fixed camera center is a special case that still gives a homography.

5. One H fits one plane

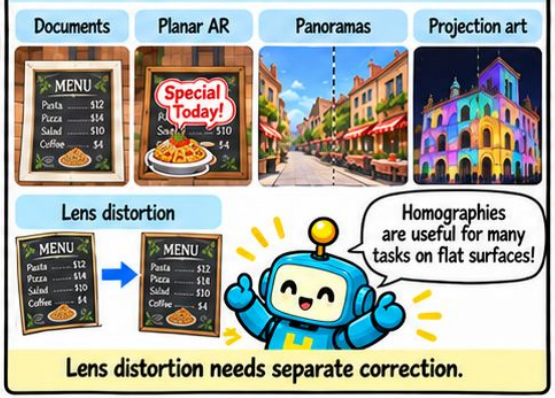


6. Depth breaks the rule



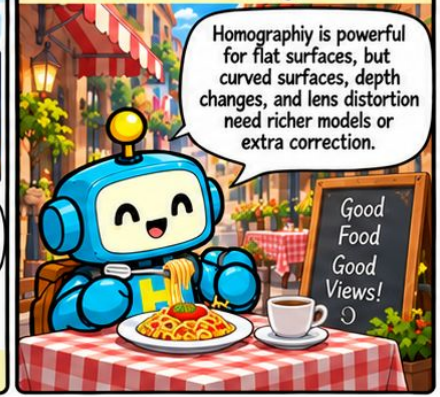
A single homography can align one plane (the menu), but it cannot correctly align objects at different depths.

7. Put it to work



Lens distortion needs separate correction.

8. Choose the right tool



PERSPECTIVE HOMOGRAPHY

WHEN MATCHES GET MESSY

9 / Advanced: RANSAC, DLT, and refinement

1 After lunch...



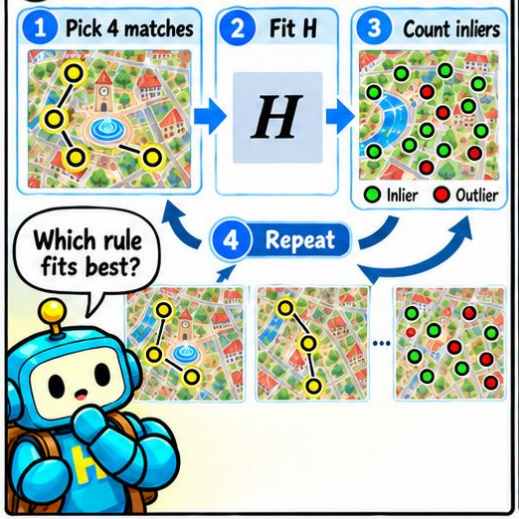
2 Uh-oh, my map!



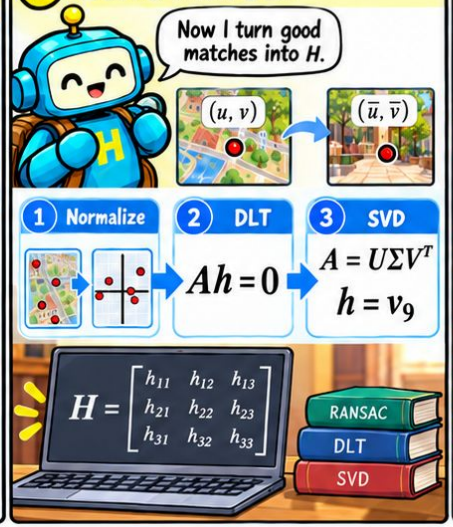
3 Detect and match



4 RANSAC tests hypotheses



5 Solve with DLT



6 Refine and head back



PERSPECTIVE HOMOGRAPHY

BEYOND THE FLAT MAP

10 / Advanced: projective geometry and cameras

1. Back to the square



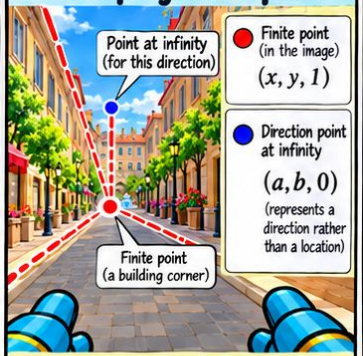
After a full day exploring, I'm back at the fountain plaza and clock tower!

2. Parallel lines meet



In the image, parallel directions can meet at a vanishing point.

3. The projective plane



Finite point (in the image) $(x, y, 1)$

Direction point at infinity $(a, b, 0)$ (represents a direction rather than a location)

The projective plane adds points at infinity so parallel directions have a place to meet.

4. Points and lines transform

Point transform (up to scale)

$$p' \sim Hp$$



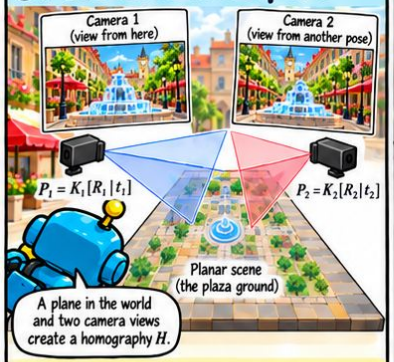
Line transform (up to scale)

$$l' \sim H^{-T} l$$



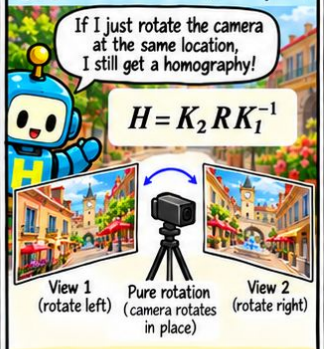
Homography transforms points and lines differently while preserving incidence.

5. Cameras and a plane



For planar scenes, camera motion and plane geometry determine the mapping.

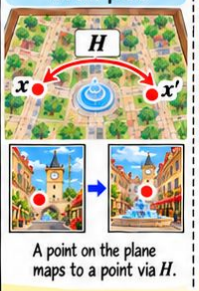
6. Pure rotation (same place)



With pure rotation, the scene is mapped by a homography: $H = K_2 R K_1^{-1}$.

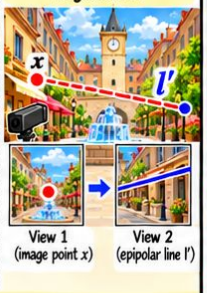
7. H versus F

H: point to point on one plane



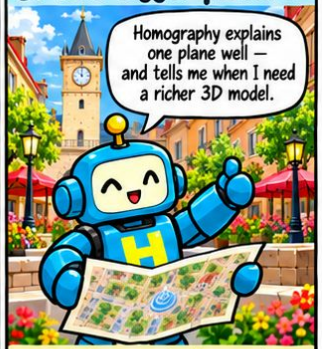
A point on the plane maps to a point via H.

F: point to epipolar line in general 3D



H explains planar correspondences; F captures general two-view geometry.

8. The bigger picture



Projective geometry, homography, and camera models help me see, map, and understand the world.

References

- **Comic pages 3–6, 9–10:** Projective transformations, geometric invariants, homogeneous coordinates, homography estimation, DLT, camera geometry, and the relationship between homography and the fundamental matrix. Hartley, R., & Zisserman, A. (2004). *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.https://doi.org/10.1017/CBO9780511811685
- **Comic pages 1–3, 5–9:** Perspective image formation, geometric transformations, image warping and rectification, feature matching, panorama stitching, and practical computer-vision applications. Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd ed.). Springer.https://doi.org/10.1007/978-3-030-34372-9
- **Comic pages 5 and 9:** Feature matching, point correspondences, inliers and outliers, and robust homography estimation using RANSAC.OpenCV. (n.d.). *Feature matching + homography to find objects*. OpenCV Documentation.https://docs.opencv.org/4.13.0/d1/de0/tutorial_py_feature_homography.html
- **Comic pages 6–8, 10:** Homography estimation, perspective transformation, camera pose, homography decomposition, and the pure-rotation case.OpenCV. (n.d.). *Basic concepts of the homography explained with code*. OpenCV Documentation.https://docs.opencv.org/4.10.0/d9/dab/tutorial_homography.html